REPORT

ON

SOLAR SYSTEM

AS A

BACK UP FOR OUR

TRAFFIC LIGHTNING SYSTEM

AUGUST, 2015.

ABSTRACT

Traffic lights form an indispensable part of our roads networks but unfortunately, it is not given the needed relevance due to mainly lack of electricity to power these traffic lights, low culture of maintenance and also ignorance on the consequence of it ineffectiveness.

This research seeks to provide engineering solutions to problems associated with the use of traffic lights connected to the grid (service lines) in my community, Kotobabi municipal town, a suburb of Accra. It used organised questionnaires to obtain the required data and also employed the use of secondary data sources to help with problem analysis, suggestions of solutions to these problems and the drawing of relevant conclusions and recommendations.

Clearly, it could be noticed that there was high level of ignorance on the effects that could envisage from the non-functioning of the grid connected traffic lights and also the absence of traffic lights at various road intersections. As a result, the creation of awareness on the consequences of this problem should to be undertaken. Solar systems should be installed as a backup for these grid connected traffic lights. Finally the Road and Safety Section of the Ghana Highway Authority should re-equip their Maintenance team and see to it that they constantly carry out routine maintenance works on these traffic lights.

ACKNOWLEDGEMENT

The success of this study required the help of various individuals. Without them, this work could not have met its objectives. I want to extend my gratitude to the following people for their invaluable help and support.

First to Jesus Christ, our Lord And Saviour, for giving me the wisdom, strength, support and knowledge in exploring things; for the guidance in helping surpass all the trials that I encountered and the determination to make this study possible.

To the staffs of ECG, who helped in getting all the necessary data and information that enabled the completion of this project work.

To Mrs. Joyce Agbeka, a staff of Ghana Highway Authority and to all other staff members at especially the Road and Safety Division, for the depth they added to my work and also broadening my understanding of the real situation at hand.

Also to all family members and friends for their endless supports and encouragements.

Finally I would like to acknowledge the KNUST Engineering Society team, for granting me the opportunity to apply my knowledge in identifying and recommending solutions to societal problems.

A big THANK YOU and May the merciful Lord bless you all.

KNOWLEDGE IN EXPLORING THINGS; FOR THE GUIDANCE IS

ENCOUNTERED AND FOR GIVING

THEIR STUDIES AND TO MAKE THIS STUDY POSSIBLE

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CHAPTER ONE
INTRODUCTION

1.1 Course Background

Engineering in society (CENGL 291) is a course that seeks students to identify societal problems and possibly probe into why and the solutions to these problems. The main objective of this course is to equip students with the requisite skills and knowledge to solving societal problems and also appreciate the need for engineering.

The course inculcates in students the habit of being sensitive to societal problems, thus to help them undertake developmental projects. As a result of the numerous problems facing the communities which have not been attended to, the management of the College of engineering thus made the input and developed the course to establish students in the field with the ability to identify and generate practical solution to them.

1.2 Objectives

The objectives of this program is to:

- ❖ Identify problems in the community that require electrical attention
- ❖ Concentrate on the problem that requires immediate attention.
- ❖ Use electrical engineering solutions to suggest on how the problem can be tackled.

1.3 Outline of the Project Report

The project report consists of six sub-chapters whose descriptions are as outlined below.

Chapter 1: This chapter gives the background description of the Engineering in society (CENGL 291) course in order for readers to understand the importance of the topic. It also seeks to outline the various specific objectives of the project.

Chapter 2: Here the research approach and design, the methods adopted, have all been explained. The means of acquiring the required data have also been outlined in this chapter. Also, a detailed description of how the map of the selected studied community/locality was prepared and furthermore, the identified problems that necessitated this project and how it was identified.

Chapter 3: Primary and secondary data acquired during the period of this study were presented, analyzed and the results were presented in this chapter. These were done within the constraints of available data. In this chapter also, the community is described and the map is also placed here.

Chapter 4: The drawing of relevant conclusion, having the research questions in focus and the making of appropriate recommendations for the project could be found outlined in this chapter.

Chapter 5: This chapter contains the list of previous research works cited (references) in this report.

Chapter 6: This contains the appendix of the write-up which comprises mainly, of samples of formulated questionnaires used in this study.

CHAPTER TWO
METHODOLOGY

2.1 Problem Identification

The problem came about when two people, a young girl around the age of 12 years and a man who looks to be in his early 30's, were knocked down by vehicles on two different roads in the locality. After probing into how the incidents occurred, it was discovered that at both instance there was a traffic light that was not functioning due to power outage. Surely one can conclude that the non-functioning of the traffic light due to erratic power supply may have contributed to the accidents. Also residents in my locality have for some time complained of vehicular traffic on some roads networks with this factor playing a contributing role. Being a committed resident in the locality, I decided to undertake this case study in order to be able to suggest possible long term measures to this menace.

2.2 Preparation of Map

First a search query 'Kotobabi Accra' was entered into the search engine of Google Map. I snapped the entire portion of the area of study with the help of the Snipping tool from Google Map. Demarcations for the boundary of study area was already done. Finally the said map was copied and pasted in the report.

2.3 Research Design/ Methodology

The adapted research approach for this study includes the use of questionnaires, in a survey, to gather primary data. The questionnaire was designed after careful consideration of the causes and effects of the problem on the livelihood of residents. The research was carried out by observation, collection and presentation of data, analyses of results and drawing of conclusion through experimental evidence. The analysed data is carefully elaborated to provide an in-depth understanding of the problem at hand.

2.4 Data Collection

Data is simply facts or information used usually to calculate, analyze, or plan something. (http://www.merriam-webster.com/dictionary/data). Both primary and secondary data were used in the analysis during this research study. An in-depth explanation of the methods used for data collection are presented below:

2.4.1 Primary Data

These are facts that the researcher collects specifically for the purpose of the research project. It was accomplished through methods such as the use of questionnaires, interviews and direct observation. One of the two questionnaires was handed out to some people within the chosen community and the other, to senior staffs of the Roads and Safety Section of the Ghana Highway Authority (GHA). And for the electrical engineering professionals at the Electricity Company of Ghana (ECG) Accra-East District Office, only interviews were conducted. Information retrieved provided an insight into the issue.

2.4.2 Secondary Data

Secondary data is information that has been collected for a purpose other than your current research project but has some relevance and utility for your research. At certain points in this study secondary data was employed through internet searches and literature reviews, all providing information relating to the different types of traffic lights.

2.5 The Questionnaire

A **questionnaire** is a research tool consisting of a series of questions and other prompts for the purpose of gathering information from respondents. Experimental findings were obtained particularly by the administration of the questionnaires and conducting of interviews. Two different sets of questionnaires were designed considering the causes and effects of the problem on the livelihood of the residents. One set of the questionnaire was designed purposefully for the residents in the locality (Kotobabi) and the other set for the staff members of the Roads and Safety Section of the Ghana Highway Authority.

The questionnaires were designed using Microsoft word and printed afterwards after which it was handed to respondents. The questionnaire played a major role in the collection of data since it did not require as much effort from the investigator as compared to verbal surveys.

In all 18 respondents were obtained at Kotobabi and 1 respondent from GHA. The questionnaire was also limited by the fact that a few of respondents were illiterates. So in such cases the content of the questionnaires was interpreted and the respondents were further guided in presenting their responses. All these were done to attain true and unbiased responses.

The investigator used a mixed question format i.e. the open and close question format to allow respondents express their views when necessary and were also limited to YES or NO alternative. The questions were developed strategically to serve as a test of the project objectives and were reviewed to prevent the repetition of questions, remove unnecessary questions and also avoid any other problem related to this research study. Copies of the used questionnaires can be found in Chapter 6.

2.6 Population and Delimitation

Attention was given to a specific population who were deemed eligible to have enough information for the topic. It comprised of a handful of residents in the chosen area, a number of workers at both the ECG and GHA. Other information were derived from the internet and the reports relating to this subject matter.

Respondents for the questionnaires were drawn from only Pigfarm, Abavana and Ebony localities all in the Kotobabi municipal area, and a suburb of Accra. This is due to the large environment I was exposed to, coupled with financial constraints and the less amount of time available to conduct this research. As a result the opinions expressed in this work might not be the precise reflection of the 10 thousands of residents in Kotobabi.

2.7 Sampling

Sample information were gathered from a few residents (Pigfarm, Abavana and Ebony) out of the over 20 thousand people in the chosen area due to reasons already stated. A total of 40 houses and 5 employees of ECG and GHA consent was to be sought about the issue at hand. It is worth noting that the researcher only interviewed the ECG staffs, they did not have to use the **first questionnaire** since most of the questions in it was not related to their work. In a nutshell, sample info for the study were gathered from:

I. Staff members of ECG and GHA

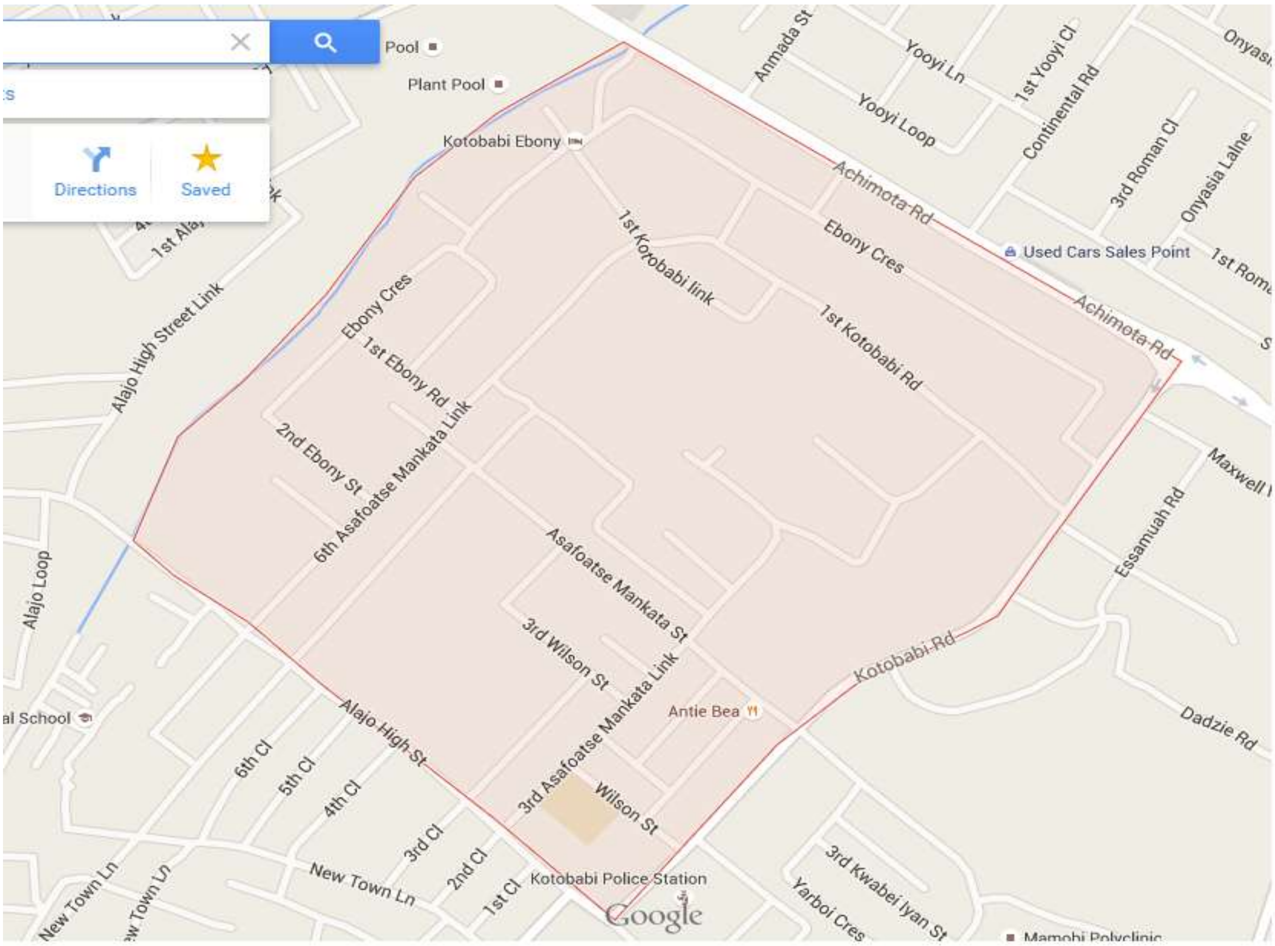
II. Selected houses in the locality

CHAPTER THREE
DATA PRESENTATION, ANALYSIS AND DISCUSSION OF RESULTS

3.1 Community Description

Kotobabi is a municipal area that is situated in the Accra –East District, a part of the Ayawaso Central Constituency. Most of the settlements are not well planned. Majority section of its populace are in to medium scale trading, with others engaged in small businesses. Institutions such as schools, hospitals, banks and others can be found in the area. The town is located at longitude **-0°12'20.59"** and latitude **5°35'48.27"** of Accra.

Map of Study Area- Kotobabi Municipal Area



Source: Google Map

3.2 Program Description

Electrical Engineering is a branch of engineering that constitutes the study and application of electricity, electronics and electromagnetism. Its subdisciplines include electronics, power engineering, telecommunications, control systems, signal processing, instrumentation, job prospects.

a. Power Engineering deals with the generation, transmission and distribution of electricity. These include transformers, electric generators, electric motors, high voltage engineering and power electronics. In Ghana, this aspect of electrical engineering is common with the Volta River Authority (VRA), Ghana Grid Company (GRIDCO) and Northern Electricity Distribution Company (NEDCO) and Electricity Company of Ghana (ECG) responsible for the generation, transmission and distribution of power respectively.
b. Control Engineering focuses on the modelling and design of dynamic systems and controllers that will cause the system to behave in a desired manner. Control Electrical Engineers achieve this with the use of electrical circuits, digital signal processors, microcontrollers, etc.
c. Electronics involves the design and testing of electronic circuits that use the properties of components such as resistors, capacitors, diodes, inductors and transistors to achieve a particular aim.
d. Signal processing deals with the analysis and manipulation of analog signals (signals that vary continuously) and digital signals (signals that vary according to a series of discrete values representing information)
e. Telecommunication focuses on the transmission of information across a channel such as an optical fibre, coax cable or free space.

- **Knowledge concerning the manufacture of the various components of a traffic light and its operation can be traced to different subdisciplines of electrical engineering. As a result, the problem identified is electrical related.**

3.3 Presentation of Survey Findings

Remember the views of respondents of the questionnaire have been interweaved with secondary data (when necessary) and analysed for the purpose of elaborating on discussions. A total of 40 questionnaires were sent to targeted residents out of which 18 responses were received. To verify whether the problem the researcher identified was indeed a real challenge to the residents in the locality and also to seek their concerns on the issue, the following questions were included in the **second questionnaire**:

- What do you think of the current traffic situation in our community?
- What may be the cause of the sudden increase in cases of vehicular and pedestrian accidents in the community?
- What of those traffic lights that do not function regularly, what may be the cause?
- Is the erratic power supply a major cause?
- A grid connected traffic light or solar powered one, which one would you welcome and why?
- What danger do you face using such roads?
- Have you made efforts to contact the right authorities to see to it that these neglected roads have these traffic lights fixed? And what was their response?

The responses received are as follows:

First, 12 of the respondents were male and 6 women; all above the age of 21 years. **All the respondents** were content with the rate of vehicle flow on the various roads in the locality. They further indicated that the roads (especially the Olusegun Obasanjo Highway) become choked with vehicles whenever the traffic lights are not functioning. PICTURES OF SUCH INSTANCES ARE PROVIDED BELOW…………………………………………….

Vehicular traffic due to non-functioning signal.

Non-functioning signal due to power cut.

Not surprising, **all the respondents** blamed the non-functioning of some of the traffic lights in the community on the **erratic power supply** that has plagued the country for some time now. In addition, **5 of the respondents** listed **low maintenance culture** on these traffic signals as also a cause of this problem. **One resident** in a short interview stressed that **high level of ignorance on the consequence** of the non-functioning of these traffic lights was one of the major cause of this problem. He argued that ignorance has subsequently led to neglect on the part of residents to put pressure on authorities so that they (authorities) can find an alternative way for these traffic signals to function. As a result, the authorities in charge also feel reluctant to address the issue. His arguments were strengthened when **80% of the respondents** answered (questionnaire) that they have not made any efforts to contact authorities to ensure the situation is given serious attention. The remaining **20%** said their calls have not been given the necessary attention. The effects of this traffic light issue as stated by the respondents are as follows:

- Minimize in business activities due to long hours spent on choked roads
- Increase in vehicular and pedestrian accidents in locality
- Lawlessness on these roads
- Waste of personal time

The respondents were quizzed on whether they would consider a solar traffic light as a solution to the problem or the current (traffic lights that are powered from the grid) ones should be maintained. Result are tabulated below.

Respondent's view	**Result (%)**
In support of solar traffic light	90
Maintenance of existing traffic ones	10

This section of the research study was necessary since the whole exercise depended on the results obtained because it served as a way of testing my solution to the problem. The issue here was that those in support of a solar traffic light argued that even though the solar powered traffic signals are expensive, their cost is still justified considering the fuel wasted in vehicular traffics and time wasted: all affecting productivity. The other section wanted no change of the traffic lights but regular maintenance works should be carried out with the hope that the energy crisis would be resolved one day due to high cost of the solar traffic lights. Further inefficiencies on our roads were uncovered during the research, the researcher discovered that still some major road intersections lacked the presence of traffic lights which was a risky activity since it would also present dangers as already stated above.

To seek clarification on the problem, the researcher was not hesitant to contact some authorities whose work field are related to the issue. At the GHA (Head Office), 3 staff members were contacted using the **first questionnaire** and just one response was obtained. The respondent was a **SUPERVISOR** at the Roads and Safety Section. The following questions in the questionnaires helped verify the concerns of the residents and also in community:

- What are the major causes of non-functioning of traffic lights on our municipal roads?
- Can we also blame it on erratic power supply**?**
- Why have we as a country been over-reliant on traffic lights connected to the grid?
- What is the life span of these grid connected traffic lights?
- Has your outfit considered solar traffic lights to be a solution to this issue?
- What will you say is the cost of installing **a single solar traffic light?**
- Should traffic lights connected to the grid be abolished?
- Does solar traffic lights pose any danger to our society?
- Can you tell me the impact of this problem of non-functioning traffic lights on our roads have on our daily activities and socio-economic development as a country?

The respondent responses are presented below:

- ➢ On the cause of the non-functioning of traffic lights, he listed a number of factors contributing to that effect. Some were:
  - ❖ Unreliable power supply.
  - ❖ Petty faults that develop in the system.
  - ❖ Deterioration of cables used for underground wiring connection with time.
  - ❖ Poor maintenance.
  - ❖ Use of substandard components.
  - ❖ Vandalism

She explained that the traffic lights positioned at various road intersections are connected by way of underground connections to the nearest service line. So any time power supply is interrupted, the traffic lights go off. Therefore the situation was out of their control. For the faults, she said they were natural because any machine or system cannot have 100% efficiency.

Concerning the cable deterioration, she made reference to the cable insulation. 'These cable insulations even though they are properly prepared so as to serve their purpose (mechanical support and moisture control), they still have a life span'. Another revelation was that, they have desisted from the use of copper cables due to theft cases previously recorded. The copper cables are very expensive than the aluminium ones. Other causes listed were use of substandard components, vandalism and poor maintenance.

- She said Ghana, over the years, has adopted a policy of installing traffic lights that are powered from the electricity (service lines). This is due to insufficient budgetary allocation to support the solar ones because of its high initial cost and therefore the cost of installing a solar traffic light is €24,000.00 **A breakdown of the materials needed when installing a solar traffic light and their respective cost is provided below:**

Car traffic light with 3 heads	**€650**
Pedestrian traffic light with 3 heads	**€850**
3.6 Pole	**€180**
Traffic controller with battery complete	**€17,700**
Cables complete	**€3,500**
1 Solar panel (150 watts)	**€1,300**
Additional panel	**€1,200**

*The life span of the panel and battery is 25 years and 5-6 years respectively.

- Another point raised was the lack of technology to develop our own solar powered traffic lights. Even though they are not the best we have no choice but to adjust to it for the time being due to financial constraint. But she stated categorically that the country has started introducing the solar powered traffic lights. Solar traffic lights may be the best solution taking account is numerous advantages like:

- ❖ It is environmental friendly and maintains a balance ecosystem by the harnessing sun rays for power production and being pollution free.
- ❖ Always provides constant power supply since it is renewable.
- ❖ No cable need for connection, reducing cost and possible deterioration of these cables.

- ➢ Finally the respondent admitted the short falls the nation has faced due to the non-functioning of the traffic signals. Vehicular and pedestrian traffics across the country has indirectly contributed to the dwindling nature of the economy. Looking at the congestion on our roads whenever there is power cut has led to minimize in business activities and other factors has played a part in our current economic deficit. Other negative effects were increase in conflicts at intersections, rise in indiscipline at intersections and loss of life and property as a result of accidents at intersections.

The researcher paid a visit to the Accra-East ECG District Office located directly opposite the Accra Girls Senior High School. Fortunately I had the chance interact (interviews) with 4 personnel with their professional background ranging from electrical technicians to electrical engineers. The details of the information gathered during our interview are provided below:

- The traffic lights and even the street lights are powered from nearest service line.
- They are connected to all the three phases to ensure constant power supply even if any phase goes off.
- About the power crisis which has led to the non-functioning of most traffic lights in the locality and the country at large, they complained of how Ghanaians blame them totally for being behind the whole situation. They sympathized with the level of loss to many businesses and even lives that have been lost all due to this situation. But they maintained their stance that even though they are the last grid company in terms of power generation, transmission and distribution in Ghana, they should not be totally blamed for the power crisis. 'WE DISTRIBUTE WHAT WE ARE SUPPLIED' said one interviewee.
- As already stated, power generation is the responsibility of Volta River Authority (VRA), power transmission is done by the Ghana Grid Company and the Electricity Company of Ghana (ECG) and Northern Electricity Distribution

Company for distribution (NEDCO). Over the years, VRA has not been able to meet the energy demands of the country as the nation continues to work hard so as to attain the status of a full medium income country by expanding the various sectors of the economy especially production . The table below provides the various power generation facilities across the country with their corresponding installed capacities and current generation capacities.

GENERATION FACILITY	**INSTALLED CAPACITY (MW)**	**CURRRENT PRODUCTION (MW)**
Akosombo Hydroelectric Power Plant	1,020	500
Kpong Hydroelectric Power Plant	160	120
Takoradi Thermal Power Station (T1)	330	150
Takoradi Thermal Power Plant (T3)	132	0
Takoradi International Company (TICO/T2)	220	100
Tema Thermal 1 Power Plant	110	60
Tema Thermal 2 Power Plant	50	0.88

Mines Reserve Power Plant	80	2.24
Solar Power Plant	2.5	0.0225
Sunon-Asogli Power Plant (SAPP)*	200	86
CENIT*	126	5.5818
Bui Hydroelectric Power Plant	400	0

Source: (P.Y.OKYERE, the Challenges of Power Generation and Distribution and its Effect on the Economy, May, 2015)

Ghana has over the years been over reliant on the Akosombo Hydroelectric Power Plant which served a total load of 100MW during its inauguration. But due to load increase and the Power Plant not being able to produce its installed capacity, the country has been hit by its biggest power crisis since independence. Other generation facilities like the Thermal Plants already listed were built to help solve this power deficit. Unfortunately their existence has rather worsen the whole situation due to sometimes shortage of gas to power the plants. Even taking a close look at the country's energy demand of 2500MW and a total installed capacity of 2846.5MW, it is clear our power reserve is around 346.5MW which is minute. Meaning the country must shed some power whenever a major fault is encountered at any of the generation stations, in order to save the whole system. And that is the situation we have found ourselves (as a country) in today.

3.4 Discussion of Solution

After fairly analyzing the discoveries made during data collection and the whole problem, the researcher is now obliged to suggest an engineering solution to the issue using knowledge from his chosen field of engineering (electrical).

3.4.1 Facts about Traffic Lightning Systems

- Traffic lights also known as traffic signals or traffic control signals are signalling device that are positioned at road intersections, pedestrian crossings and other places to control the movement of vehicles in a desired manner usually through timing. Traffic lights alternate the right of way accorded to road users by displaying lights of a standard colour (red, yellow, and green) following a universal colour code.
- They come in various types based on their function and positioning. Common ones are the single aspect, dual aspect, three or more aspects and pedestrian or cyclist crossing lights. Aspect is the visual appearance of the signal.
    - Single aspect is the simplest traffic light and comprises either a single or a pair of coloured aspects that warns any user of the shared right of way of a possible conflict or danger.
    - The double aspects are often seen at railway crossings, fire stations, and at intersections of streets. They flash yellow when cross traffic is not expected, and turn red to stop traffic when cross traffic occurs.
    - Three aspect is the standard red light above the green, with yellow between.

Images of the various types are shown on the next page:

SINGLE ASPECT **DUAL ASPECT** **THREE ASPECTS**

Source: Google

- The Vehicle and Pedestrian Signal Head Indications have their meanings:

I. Red Signal Head Indications

This means all vehicles or pedestrians approaching the intersection should stop before the cross walk marked on the asphalt.

II. Green Signal Head Indications

The green light means "go", that is, all vehicles and pedestrians who stopped or are approaching the intersection can move.

III. Yellow/Amber Signal Head Indications

In Ghana, amber light means vehicles should get ready to stop if able to do so safely. Flashing amber means, traffic light intersection is faulty or the system is resting. The second case takes place normally from 11:00 pm to 5:00 am.

- Traffic lights is basically made up of the signal lights connected to a controller. Traffic lights are computer controlled. The controller is in a box near the lights. The controller is programmed according to the type of junction with the aid of a controller handset or laptop. Signal heads mounted on poles are connected by a three- phase four wire connection to a controller cabinet. Power is tapped from the nearest ECG service pole to the traffic light controller. The controller runs on single phase but it is normally connected to three phase four wire power so that it can change to another phase in case of phase failure.

➢ Traffic lights have three working modes:

i. Timed Mode of Operation
ii. Detection Mode of Operation
iii. Timed and Detection Mode of Operation.

3.5.2 Mode of Operation of a Solar System

➢ The solar system uses chemical and physical processes to convert photons to electric current when sun rays strike a suitable semiconductor device. It comprises solar panels and solar cells.
➢ Photons in sunlight hit the solar panel and are absorbed by semi-conducting materials.
➢ Electrons (negatively charged) are knocked loose from their atoms as they are excited. Due to the special chemical composition of solar cells, the electrons are only allowed to move in a single direction. The chemical bonds of the material are vital for the process to work, and often silicon, bonded with boron or phosphorus is used in different layers.
➢ An array of solar cells converts solar energy into a usable amount of direct current (DC) electricity.

A SOLAR SYSTEM

Source: Google

3.5.3 Proposed Solution

The theme for the research '**SOLAR SYSTEM AS A BACK UP FOR OUR TRAFFIC LIGHTNING SYSTEM'** gives a first sight overview of the whole study. The solution is so simple, it entailed the modification of the current traffic lighting system by including a solar system to provide power when the main power supply drops or goes off by the energization of a relay. The proposed solution was influenced by factors like:

- The cost of installing a complete solar traffic light (**€24,000.00)**.

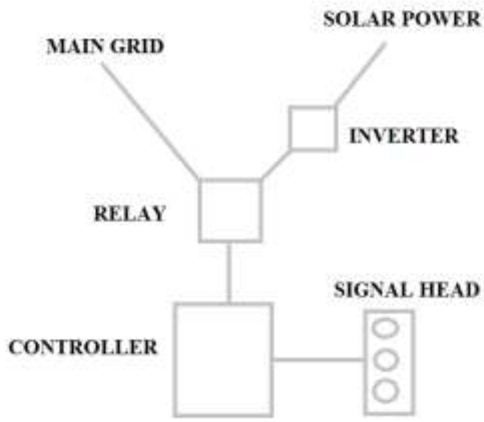
- The need for the traffic lightning system to always function.

In addition to the setup, an **inverter** with preferred rating must be added. The traffic lightning system uses an ac power for its operation but the end-product of the solar system is dc. As a result, an inverter is required to convert the dc of the solar system to ac. An illustration of how the whole setup will function is given below:

- The primary system of the setup will be made up of the old traffic lights with power supply from the next utility line (ECG).
- The solar system will be positioned above or at any suitable position. It will serve as a backup system.
- Under constant power supply to the utility line (ECG), the primary system works. But in case of any eventuality like power outage, the backup system (solar) will automatically steps in to provide power supply by the energization of a **relay**.
- This will ensure constant functioning of the traffic lights.

TRAFFIC LIGHT WITH SOLAR BACKUP SYSTEM

Source: Google



SCHEMATIC REPRESENTATION OF SOLUTION

CHAPTER FOUR CONCLUSION AND RECOMMENDATION

4.1 Conclusions

The key findings discovered during this research study are:

- The non-functioning of the traffic lightning system is due to:
  - The current power crisis and vandalism.
  - Petty faults that develop in the system and use of substandard components.
  - Deterioration of cables used for underground wiring connection with time and poor maintenance.
- Residents in the locality were ignorant on the usefulness of the traffic signals and as a result felt reluctant to report or complain of any fault or malfunctioning of these lights.
- Routine maintenance works were not undertaken on the traffic lightning system.
- Solar traffic systems are more efficient and environmentally friendly than the traffic systems powered from the grid (service line) but the country cannot totally rely on them (solar traffic light) due to its high installation cost. As a result, a combination of the two would be less expensive and the best solution to the problem.
- Residents confirmed the unavailability of traffic systems on certain major road intersections and the risk it posed to them.
- The non-functioning of these signal lights has negatively impacted our daily activities through heavy vehicular traffics on the roads.

4.2 Recommendations

Suggested measures to the problem are:

- Backing up our traffic systems with a solar system at all intersections across the nation.
- Routine maintenance works on the lights should be done.
- Creation of awareness among the public on the essence of the traffic systems and the risks they pose when they are not functioning. Also a toll free line should be acquired to help the general public report any issue of malfunctioning of these signal lights and other related grievances.
- Usage of quality components during traffic lights installation.
- Prosecuting people who vandalize traffic lights by forcing them to pay a specific fine.

CHAPTER FIVE
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CHAPTER SIX
APPENDICES

6.1 Questionnaire for Professionals

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL/ELECTRONICS ENGINEERING

SOLAR SYSTEM AS A BACK UP FOR OUR TRAFFIC LIGHTNING SYSTEM

Dennis Elorm Fleku is an undergraduate student of the above named institution who is conducting a research on the use of **SOLAR SYSTEM AS A BACK UP FOR OUR TRAFFIC LIGHTNING SYSTEM** and needs your assistance to make this research a success. Kindly cope and do not hesitate to be of help by answering the questions below:

PART A

1. Place of work: ....
2. Age: .... Gender: male ☐ female ☐
3. What are the major causes of non-functioning traffic lights on our municipal roads?
   ……………………………………………………………………………………………..
   ……………………………………………………………………………………………..
4. Can we also blame it on erratic power supply**? YES or NO**
5. Why have we as a country been over-reliant on traffic lights connected to the grid?
   ……...
   ……………………………………………………………………………………………..
6. What is the life span of these grid connected traffic lights? ....
7. Has your outfit considered solar traffic lights to be a solution to this issue? **YES or NO**
8. What will you say is the cost of installing **a single solar traffic light**? ....
9. Should traffic lights connected to the grid be abolished? **YES or NO** Why?
   ……………………………………………………………………………..............
10. Does solar traffic lights pose any danger to our society? **YES or NO**
11. Can you tell me the impact of this problem of non-functioning traffic lights on our roads have on our daily activities and socio-economic development as a country?
    ……………………………………………………………………………………………
    ……………………………………………………………………………………………
12. What measures has your outfit put in place to resolve this issue?
    ……………………………………...
    ……………………………………………………………………………………………
    ……………………………………………………………………………………………

6.2 Questionnaires for Residents

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL/ELECTRONIC ENGINEERING

SOLAR SYSTEM AS A BACK UP FOR OUR TRAFFIC LIGHTNING SYSTEM

Dennis Elorm Fleku is an undergraduate student of the above named institution who is conducting a research on the use of **SOLAR SYSTEM AS A BACK UP FOR OUR TRAFFIC LIGHTNING SYSTEM** at various road intersections in the locality and needs your assistance to make this research a success. Kindly cope and do not hesitate to be of help by answering the questions below:

PART A

1. Area/Location of residence: ....
2. Gender: male ☐ female ☐

3. What do you think of the current traffic situation in our community? ....
…………………………………………………………………………………………………...
4. A. What of those traffic lights that do not function regularly, what may be the cause?
....
B. Is the erratic power supply a major cause? **YES or NO**
C. A grid connected traffic light or solar powered one, which one would you welcome and why?
…………………………………………………………………………………………………

5. What danger do you face using such roads? ....

6. A. Have you made efforts to contact the right authorities to see to it that these neglected roads have these traffic lights fixed? **YES or NO**
B. And what was their response? ……………………………………………………………………..

7. What may be the cause of the sudden increase in cases of vehicular and pedestrian accidents in the community? ....

8. Are you satisfied with the unavailability of traffic lights at certain road intersections in the locality like Pigfarm, Abavana junctions and others? **YES or NO**

9. Do you feel your community has been neglected? **YES or NO**